

DUKI, Russian Federation

WMO: 314820

Lat: 51.717N

Lon: 135.933E

Elev: 132

StdP: 99.75

Time Zone: 10.00 (E10)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.8	(c) -38.8	(d) -42.6	(e) 0.1	(f) -40.4	(g) -40.7	(h) 0.1	(i) -38.7	(j) 8.8	(k) -13.6	(l) 7.6	(m) -11.9	(n) 0.0	(o) N/A	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.4	(c) 30.3	(d) 21.1	(e) 28.3	(f) 20.3	(g) 26.3	(h) 19.3	(i) 22.8	(j) 27.8	(k) 21.5	(l) 26.5	(m) 20.3	(n) 25.1	(o) 1.5	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	16.0	26.2	19.7	14.7	24.4	18.5	13.6	23.0	68.4	27.8	63.4	26.6	59.2	25.5	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.1	(b) 4.8	(c) 3.8	(e) -43.4	(f) 33.3	(g) 3.1	(h) 1.9	(i) -45.7	(j) 34.7	(k) -47.5	(l) 35.7	(m) -49.2	(n) 36.8	(o) -51.5	(p) 38.1		
			DB													
			WB													
				(e) -43.2	(f) 25.3	(g) 2.7	(h) 2.1	(i) -45.1	(j) 26.8	(k) -46.7	(l) 28.0	(m) -48.2	(n) 29.2	(o) -50.1	(p) 30.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-26.9	-20.3	-11.4	0.5	7.8	14.9	18.5	17.0	10.1	0.9	-12.8	-23.8		
	DBStd	16.47	6.05	6.26	5.80	4.38	4.37	4.28	3.40	3.00	3.56	4.82	6.06	6.12		
	HDD10.0	5103	1143	848	662	286	99	8	0	0	43	282	684	1048		
	HDD18.3	7516	1401	1082	921	535	328	118	41	63	248	540	934	1306		
	CDD10.0	712	0	0	0	1	30	156	263	217	45	1	0	0		
	CDD18.3	83	0	0	0	0	1	16	46	21	0	0	0	0		
	CDH23.3	1210	0	0	0	0	44	332	586	242	7	0	0	0		
	CDH26.7	362	0	0	0	0	7	102	182	71	1	0	0	0		
(14) Wind	WSAvg	1.0	0.6	0.9	1.2	1.5	1.5	1.1	1.0	0.9	1.0	1.2	0.9	0.5		
(15) Precipitation	PrecAvg	646	19	14	22	43	77	81	112	108	71	50	27	25		
	PrecMax	816	57	48	61	95	144	145	219	234	136	138	60	86		
	PrecMin	389	2	4	2	14	23	23	1	38	18	16	5	3		
	PrecStd	114	14	9	15	22	33	29	49	51	33	31	17	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.6	-1.7	10.2	18.6	28.4	31.9	32.7	32.0	24.4	17.6	3.2	-4.0		
		MCWB	-6.3	-3.9	4.6	9.2	15.3	21.0	22.2	23.0	16.4	9.4	-0.6	-5.4		
	2%	DB	-7.9	-4.8	5.5	14.6	24.7	29.5	30.6	28.6	21.8	14.5	0.5	-7.2		
		MCWB	-8.6	-6.0	0.9	6.9	14.3	20.0	21.6	21.1	15.6	7.9	-1.8	-8.6		
	5%	DB	-11.7	-7.0	1.8	11.4	21.6	27.2	28.7	26.0	19.4	11.5	-1.3	-10.4		
		MCWB	-12.4	-8.1	-1.4	5.2	12.9	18.6	21.2	19.9	14.5	6.0	-3.7	-11.2		
	10%	DB	-16.4	-9.6	-0.7	8.6	18.3	24.8	26.9	23.7	16.8	9.0	-3.3	-13.0		
		MCWB	-17.0	-10.8	-3.2	3.4	11.0	17.4	20.4	18.6	12.9	4.6	-5.0	-13.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.2	-3.5	4.9	9.6	18.1	24.4	24.7	24.4	18.7	10.4	0.5	-5.3		
		MCDB	-5.9	-2.3	10.3	16.9	26.0	30.7	30.1	29.6	22.2	15.6	2.2	-4.4		
	2%	WB	-8.5	-5.9	1.1	7.2	15.5	21.2	23.0	22.1	16.7	8.2	-1.5	-8.6		
		MCDB	-7.8	-5.0	5.2	13.8	22.2	27.5	28.0	26.7	20.5	13.1	0.6	-7.4		
	5%	WB	-12.5	-8.2	-0.9	5.4	13.4	19.7	22.0	20.7	15.1	6.6	-3.5	-11.3		
		MCDB	-11.6	-7.0	1.9	11.0	20.4	25.4	27.2	24.6	17.9	10.9	-1.8	-10.5		
	10%	WB	-17.0	-10.8	-3.4	3.7	11.6	18.1	20.9	19.5	13.8	5.0	-4.9	-13.6		
		MCDB	-16.4	-9.7	-0.8	8.0	18.0	22.9	25.9	23.1	16.1	8.4	-3.2	-13.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	16.8	16.7	11.9	12.8	14.0	12.4	10.7	11.1	11.5	12.6	13.1		
		MCDBR	13.6	16.1	17.4	17.6	20.6	19.1	17.4	16.4	16.1	16.6	12.7	12.4		
		MCWBR	12.6	14.2	13.2	10.5	11.2	10.3	9.7	9.6	10.0	9.9	10.0	11.4		
	5% WB	MCDBR	13.2	14.7	16.0	16.8	19.2	16.6	15.3	14.3	13.3	15.7	11.3	12.0		
		MCWBR	12.2	13.0	11.8	10.3	11.3	10.0	9.5	9.5	9.0	9.7	9.1	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.298	0.369	0.480	0.562	0.503	0.502	0.424	0.377	0.382	0.314	0.265		
	taud		2.115	2.054	1.897	1.790	1.757	2.012	2.064	2.301	2.358	2.135	2.087	2.085		
	Ebn at Noon		727	801	797	748	706	762	755	804	803	690	639	648		
	Edh at Noon		74	107	154	195	212	166	155	116	98	100	77	65		
(44) All-Sky Solar Radiation	RadAvg		1.48	2.67	4.29	5.21	4.95	5.42	4.95	4.32	3.39	2.13	1.41	1.13		
	RadStd		0.05	0.08	0.17	0.39	0.52	0.58	0.50	0.43	0.30	0.17	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page